Six-unit prime-norm scalar radices: evidence report

Date: 2026-10-19
Scope: the secp256k1 GLV coefficient-lattice radices `233−τ` (norm 53,593) and `190−τ` (norm 35,533), their 16-step digit selection, unit-orbit tables, and the archived Tier 1 paired batch comparisons. The comparison is hot-table throughput on a frozen scalar batch, not one-call latency or end-to-end ECDLP performance.

Question and method

Do the prime-norm radices give exact 16-position scalar recodings with smaller orbit tables, and how do the 233−τ and 190−τ tables compare with each other and the inherited 18-position rolling table on the frozen batch workload?

The integer screens exhaust the residue domains, six-unit covariance and selected 16-step terminal bounds. Lean modules prove the stated lattice identities and arithmetic bounds; candidate-specific point-group instantiation remains open as noted in the README. Independent affine replays and hash-pinned Rust overlays compare results against ordinary binary scalar multiplication. The frozen batch panels use the same 128 scalars, 16 passes per arm, alternating order, and adjacent A/A controls; their committed analysis scripts consume retained raw run/pair receipts.

Curve and subgroup: secp256k1, `EC1P256Csecp256k1ha8dc9f115a15` in `crypto:docs/curves/registry.json`; $y^2=x^3+7 \bmod p$, prime subgroup order `n = FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364` cofactor 1, standard generator `G`. Both radix rules use the same fixed τ eigenvalue and subgroup; no new curve or isogeny is introduced.

Findings

- **Derived and exhaustively checked:** the `233−τ` selector covers all 53,593 residues, with a global-minimum check over the sign-canonical domain and a 16-step terminal bound. The `190−τ` refinement covers all 35,533 residues and has the smaller point payload within the declared prime-norm sweep `180 ≤ T ≤ 233`.
- **Independently replayed:** affine and native implementations matched binary scalar multiplication on 388 scalars for each radix; the native release holdouts include all 128 frozen points. Lean arithmetic, digit, and trajectory proofs are linked in the experiment README. The complete formal secp256k1 point-group/coordinate-action bridge remains open; empirical point replay is the concrete implementation check.
- **Measured, A/A-qualified hot-table batches:** one RunPod `cpu3c` allocation, instance `thyvmdqmzsz5ik`, 8 vCPU/16 GB in EU-RO-1. The captured execution record reports x86_64 Ubuntu 24.04.4 in Docker on an AMD EPYC 9655 96-Core Processor host. The worker was pinned to CPU 13 from the exposed set `{2,13,40,74,98,109,136,170}`; one NUMA node was reported. CPU quota was unlimited, while host exclusivity and memory policy were unverified; physical memory type/speed/channel data remain unknown. Each panel completed 96 arms/48 pairs with stable source and binary artifacts and no adjacent A/A failures. Ratios are candidate/reference throughput; intervals are repeat-bootstrap 95% intervals on this one allocation:

Candidate / reference	Operations per arm	Paired median ratio	95% repeat interval	Retained bytes (reference → candidate)
190−τ / 18-position rolling	8,192	1.0492	[1.0382, 1.0608]	2,448,196 → 6,829,948
233−τ / 18-position rolling	2,048	1.0340	[1.0285, 1.0366]	2,448,196 → 9,363,076

Candidate / reference	Operations per arm	Paired median ratio	95% repeat interval	Retained bytes (reference → candidate)
190−τ / 233−τ	2,048	1.0143	[1.0054, 1.0438]	9,363,076 → 6,829,948

The first comparison also has a separate failed-gate 2,048-operation panel; the one-call latency panels failed A/A controls and have no qualified latency ratio. All failures and raw arms remain preserved in the experiment directory. Table preparation is reported separately in the README and is outside the hot-table interval. These results do not compare full scalar multiplication including setup and do not establish a cryptanalytic advantage.

Visuals and graph audit

The editable diagram source and rendered diagram show the radix/seed/fold path. The plot source reads the three committed analysis JSON files and creates the vector plot, using only A/A-qualified batch panels. The PDF report embeds both figures.

Checked `docs/BENCHMARKS.md`, `docs/perf/PERFORMANCE_INDEX.md`, `docs/perf/weights.json`, and `experiments/volcano-ic/figures/` (`census.svg/.png`, `ecdlp.svg/.png`, `tau.png`). Those canonical artifacts cover library/suite kernels or index-calculus and volcano results; these measurements are a distinct fixed-base Rust scalar-table experiment. The experiment-specific paired plot is new, but no existing canonical index row or graph is affected. The curve is already registered in the sibling crypto curve registry; no local curve or route entry changes.

Open checks and reproduction

The single-call latency ranking is unqualified, the one-allocation throughput interval is specific to this frozen workload and allocation, and the formal point-group bridge remains open. A second allocation or one-call comparison is outside this report; no new benchmark was launched for this report. Reproduce the archived analyses with the README's `analyze_batch*.py` scripts; they validate committed receipts rather than invoking the RunPod service.

Source records

- Experiment README and full claim boundaries
- Three qualified batch analyses, 233−τ versus 18-position analysis, 233−τ versus 190−τ analysis
- Raw paired receipts and manifests
- Integer screens, Lean proofs, and native overlays, Lean fresh-build script
- Source revision before this report: PR #901 head `3dd9c44e90ef16a60ed8017518b0544d3decfc97`

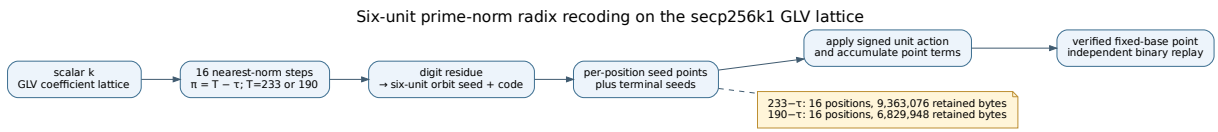


Figure 1: Six-unit prime-norm radix recoding and table fold

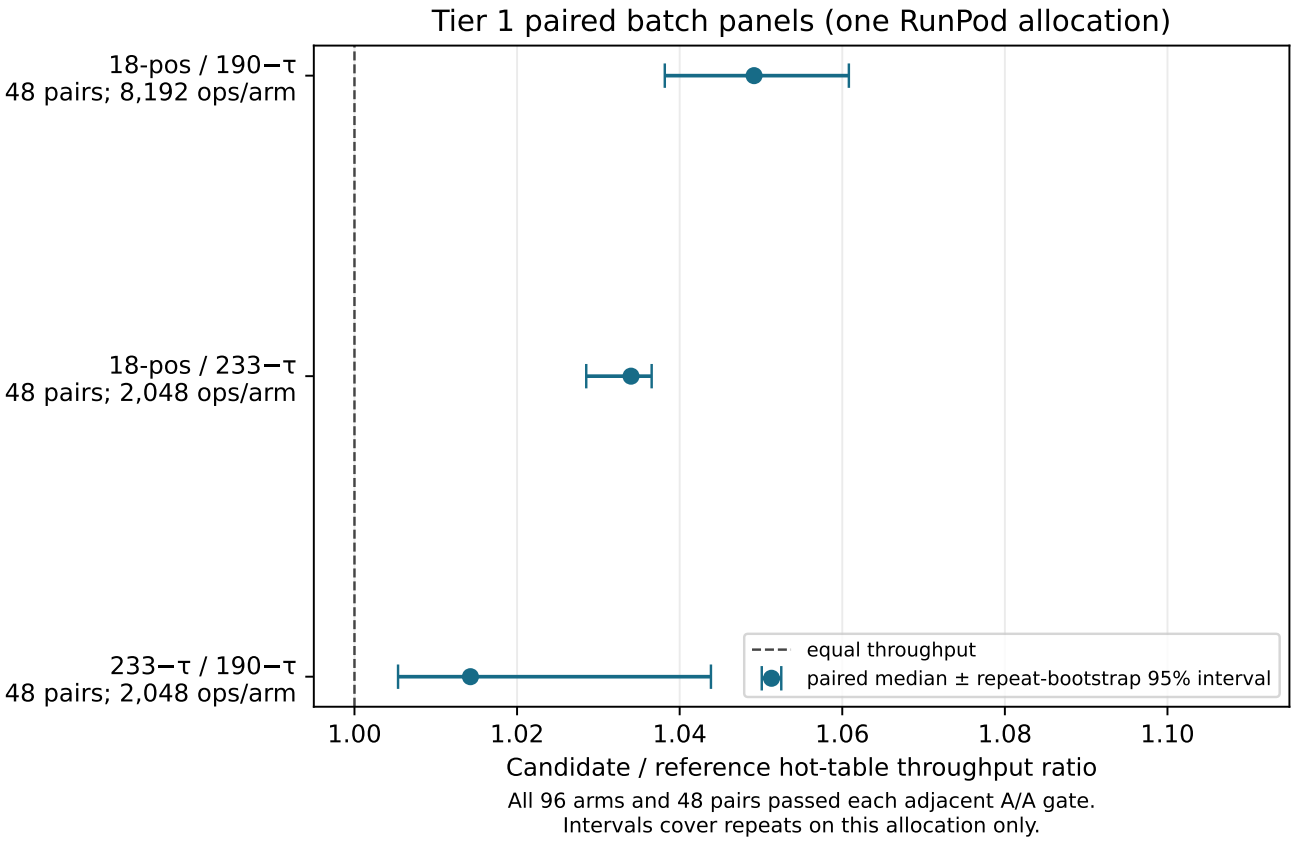


Figure 2: A/A-qualified paired batch throughput panels